Lab Assignment 4

Title: Create a Student Record Management System using linked list
• Use a singly/doubly linked list to store student data (Roll No, Name, Marks).
• Perform operations: Add, Delete, Update, Search, and Sort.
• Display records in ascending/descending order based on marks or roll number.

Program code:

```
#include <iostream>
#include <string>
using namespace std;

//node structure
struct Node {
    int rollNo;
    string name;
    float marks;
    Node* next;
};

//Head pointer
Node* head = NULL;

//Function to add a student
void addStudent() {
    Node* newNode = new Node();

    cout << "\nEnter roll no.: ";
    cin >> newNode->rollNo;

    cout << "Enter name: ";
    cin.ignore();
    getline(cin, newNode->name);

    cout << "Enter marks: ";
    cin >> newNode->marks;

    newNode->next = NULL;

    //If list is empty
    if (head == NULL) {
        head = newNode;
    } else {
```

```
        Node* temp = head;

        while (temp->next != NULL) {
            temp = temp->next;
        }
        temp->next = newNode;
    }
    cout << "\nStudent added successfully!\n";
}

//Function to display students
void displayStudents() {
    if (head == NULL) {
        cout << "\nNo student records found.\n";
        return;
    }

    Node* temp = head;

    cout << "\n--------------------------------\n";
    cout << "Roll No\t Name\t\tMarks";
    cout << "\n--------------------------------\n";

    while (temp != NULL) {
        cout << temp->rollNo << "\t "
             << temp->name << "\t\t"
             << temp->marks << endl;

        temp = temp->next;
    }
    cout << "--------------------------------\n";
}

//Function to search a student

void searchStudent() {
    int roll;
    cout << "\nEnter roll no to search: ";
    cin >> roll;

    Node* temp = head;

    while (temp != NULL) {
        if (temp->rollNo == roll) {
```

```
            cout << "\nStudent found!\n";
            cout << "Roll No: " << temp->rollNo << endl;
            cout << "Name   : " << temp->name << endl;
            cout << "Marks  : " << temp->marks << endl;
            return;
        }
        temp = temp->next;
    }
    cout << "\nStudent not found.\n";
}

//Function to delete a student
void deleteStudent() {
    int roll;

    cout << "\nEnter roll no. to delete: ";
    cin  >> roll;

    Node* temp = head;
    Node* previous = NULL;

    //search for the node
    while (temp != NULL && temp->rollNo != roll) {
        previous = temp;
        temp = temp->next;
    }

    //student not found
    if (temp ==NULL) {
        cout << "\nStudent not found.\n";
        return;
    }

    //delete first node
    if (previous == NULL) {
        head = temp->next;
    } else {
        previous->next = temp->next;
    }

    delete temp;

    cout << "\nStudent deleted successfully!\n";
}
```

```
//Function to update student
void updateStudent() {
    int roll;

    cout << "\nEnter roll no. to update: ";
    cin >> roll;

    Node* temp = head;

    while (temp != NULL) {
        if (temp->rollNo == roll) {
            cout << "Enter new name: ";
            cin.ignore();
            getline(cin, temp->name);

            cout << "Enter new marks: ";
            cin >> temp->marks;

            cout << "\nStudent record updated successfully!\n";
            return;
        }
        temp = temp->next;
    }
    cout << "\nStudent not found.\n";
}

//Function to sort students
void sortStudents() {
    if (head == NULL || head->next == NULL) {
        cout << "Not enough records to sort.\n";
        return;
    }
    int choice;

    cout << "\nSort by: \n1. Roll no. \n2. Marks";
    cout << "\nEnter choice: ";
    cin >> choice;

    int order;

    cout << "\nOrder: \n1. Ascending \n2. Descending";
    cout << "\nEnter choice: ";
    cin >> order;
```

```
    //Bubble sort
    for (Node* i = head; i != NULL; i = i->next) {
        for (Node* j = i->next; j != NULL; j = j->next) {
            bool swapRequired = false;

            if (choice == 1) {
                //Sort by roll no.
                if (order == 1 && i->rollNo > j->rollNo)
                    swapRequired = true;

                if (order == 2 && i->rollNo < j->rollNo)
                    swapRequired = true;
            } else if (choice == 2) {
                //Sort by marks
                if (order == 1 && i->marks > j->marks)
                    swapRequired = true;

                if (order == 2 && i->marks < j->marks)
                    swapRequired = true;
            }

            //Swap student data
            if (swapRequired) {
                swap(i->rollNo, j->rollNo);
                swap(i->name, j->name);
                swap(i->marks, j->marks);
            }
        }
    }
    cout << "\nRecords sorted successfully!\n";
}

//Main function
int main() {
    int choice;

    do {
        cout << "\n==================================\n";
        cout << " STUDENT RECORD MANAGEMENT SYSTEM";
        cout << "\n==================================\n";

        cout << "1. Add Student\n";
        cout << "2. Delete Student\n";
```

```
        cout << "3. Update Student\n";
        cout << "4. Search Student\n";
        cout << "5. Sort Students\n";
        cout << "6. Display Students\n";
        cout << "7. Exit\n";

        cout << "\nEnter your choice: ";
        cin >> choice;

        switch(choice) {
            case 1:
                addStudent();
                break;

            case 2:
                deleteStudent();
                break;

            case 3:
                updateStudent();
                break;

            case 4:
                searchStudent();
                break;

            case 5:
                sortStudents();
                break;

            case 6:
                displayStudents();
                break;

            case 7:
                cout << "\nThank you!\n";
                break;

            default:
                cout << "\nInvalid choice!\n";
        }
    } while(choice != 7);
    return 0;
}
```

Output:

```
==================================
 STUDENT RECORD MANAGEMENT SYSTEM
==================================
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 1

Enter roll no.: 101
Enter name: Rahul Sharma
Enter marks: 85.5

Student added successfully!

==================================
 STUDENT RECORD MANAGEMENT SYSTEM
==================================
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 1

Enter roll no.: 103
Enter name: Priya Patil
Enter marks: 92

Student added successfully!

==================================
 STUDENT RECORD MANAGEMENT SYSTEM
==================================
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 1

Enter roll no.: 102
```

Enter name: Amit Kumar
Enter marks: 78.5

Student added successfully!

==========
 STUDENT RECORD MANAGEMENT SYSTEM
==========
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 6

Roll No Name Marks
---
101 Rahul Sharma 85.5
103 Priya Patil 92
102 Amit Kumar 78.5

==========
 STUDENT RECORD MANAGEMENT SYSTEM
==========
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 4

Enter roll no to search: 103

Student found!
Roll No: 103
Name : Priya Patil
Marks : 92

==========
 STUDENT RECORD MANAGEMENT SYSTEM
==========
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students

```
6. Display Students
7. Exit

Enter your choice: 3

Enter roll no. to update: 102
Enter new name: Amit Sharma
Enter new marks: 88

Student record updated successfully!

==================================
 STUDENT RECORD MANAGEMENT SYSTEM
==================================
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 5

Sort by:
1. Roll no.
2. Marks
Enter choice: 1

Order:
1. Ascending
2. Descending
Enter choice: 1

Records sorted successfully!

==================================
 STUDENT RECORD MANAGEMENT SYSTEM
==================================
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 6

--------------------------------
Roll No  Name           Marks
--------------------------------
101      Rahul Sharma    85.5
102      Amit Sharma     88
```

```
103      Priya Patil     92
---------------------------------

==================================
 STUDENT RECORD MANAGEMENT SYSTEM
==================================
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 2

Enter roll no. to delete: 102

Student deleted successfully!

==================================
 STUDENT RECORD MANAGEMENT SYSTEM
==================================
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 6

---------------------------------
Roll No  Name            Marks
---------------------------------
101      Rahul Sharma    85.5
103      Priya Patil     92
---------------------------------

==================================
 STUDENT RECORD MANAGEMENT SYSTEM
==================================
1. Add Student
2. Delete Student
3. Update Student
4. Search Student
5. Sort Students
6. Display Students
7. Exit

Enter your choice: 7

Thank you!
```